

Power BI Calculations, DAX Basics, and Dashboard Creation

Session: W5S4

1. Recap of Previous Session

In the previous session, we covered the Power BI Desktop interface, importing datasets, and creating basic visualizations like bar and pie charts. Today, we focus on **Data Analysis Expressions (DAX)** and building a cohesive dashboard.

2. What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI to create calculated columns and measures. It is similar to Excel formulas but optimized for large, relational datasets.

DAX helps answer business questions such as:

- What are the total sales and total profit?
 - What is the year-over-year growth rate?
 - What is the average revenue per transaction?
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3. Measures in Power BI

A **Measure** is a calculation performed on aggregated data. Measures are dynamic; they change based on the filters or slicers selected on the dashboard.

Creating a Measure:

1. Go to the **Modeling** tab.
2. Click **New Measure**.
3. Enter the formula: `Total Sales = SUM(Sales[Sales])`

4. Common DAX Formulas

- **Total Profit:** `Total Profit = SUM(Sales[Profit])`
- **Average Sales:** `Average Sales = AVERAGE(Sales[Sales])`
- **Profit Margin:** `Profit Margin = DIVIDE([Total Profit], [Total Sales])`
 - Note: `DIVIDE()` is preferred as it safely handles division by zero errors.

5. Visualizing Key Metrics

Card Visualization (KPIs)

Card visuals display a single, important number. These are the first things a stakeholder looks at.

Metric	Example Value
Total Sales	150,000
Total Profit	25,000
Profit Margin	16.6%

Line Chart (Trend Analysis)

Used to see how sales or profit change over time (e.g., Sales by Month).

- **Axis:** Date or Month
 - **Values:** Sales Measure
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6. Creating an Interactive Dashboard

A professional dashboard should follow a logical layout to ensure it is easy to interpret.

Recommended Layout:

- **Top Section:** KPI Cards (Sales, Profit, Margin).
 - **Middle Section:** Sales by Product (Bar Chart) and Sales by Region (Pie Chart).
 - **Bottom Section:** Monthly Sales Trend (Line Chart).
 - **Side/Top Panel:** Slicers (Region, Year, Category) for interactivity.
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7. Adding Interactivity with Slicers

Slicers act as visual filters. When a user clicks "East Region" in a slicer, every other chart on the page updates to show only data for the East.

Steps to add a Slicer:

1. Select the **Slicer** icon from the Visualizations pane.
 2. Drag a categorical field (like **Region** or **Year**) into the Field well.
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8. Formatting and Publishing

Formatting Tips:

- **Data Labels:** Turn them on so users don't have to hover to see values.
- **Titles:** Use clear, descriptive titles (e.g., "Monthly Sales Trend" instead of "Sales by Month").
- **Colors:** Use a consistent color palette that aligns with business branding.

Publishing:

Once the report is ready, click **Publish** to upload it to the **Power BI Service (Cloud)**. This allows others to view the report in a web browser without needing Power BI Desktop.

9. Practical Exercise

Dataset Columns: Order Date, Product, Region, Sales, Profit.

Tasks:

1. Import the dataset into Power BI.
2. Create measures for **Total Sales**, **Total Profit**, and **Profit Margin**.
3. Build a dashboard containing:
 - 3 KPI Cards.
 - A Bar Chart for Sales by Product.
 - A Line Chart for Sales Trends.
 - A Slicer for Region.
4. Format the charts with proper titles and colors.

10. Learning Outcomes

By the end of this session, you will be able to perform complex business calculations using **DAX**, build **interactive dashboards**, and **publish** professional reports for organizational use.

